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1239: EMERGING TRENDS AND APPLICATIONS OF DEEP LEARNING FOR BIOMEDICAL DATA ANALYSIS

Smart client selection strategies for enhanced federated learning in digital healthcare applications

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Abstract
Federated Learning (FL) trains AI models in healthcare without sharing patient data. FL computes client models locally and combines them to create a global model. However, involving all clients is impractical due to resource limitations. Random selection of a subset of clients in each FL round can pose challenges for resource-limited devices, leading to longer processing times and potential training failures. To tackle these obstacles, this research proposes a novel strategy for FL that treats each training round as a client selection process to improve the efficiency and effectiveness of FL in healthcare applications, where data privacy is paramount. The approach begins by calculating the uncertainty value for each client, which quantifies the contribution of the client's data to the overall model. Clients are then ranked based on their uncertainty values, and those with higher loss values are given a higher probability of participating in the training process. The experimental outcomes clearly show that the proposed strategy effectively makes 1.3x training faster, and 30% lowers communication expenses, conserves computational resources, and enhances model performance when contrasted with random client selection.

Keywords Federated learning · Client selection · Digital healthcare · Edge computing · Distributed computing

1 Introduction

With each passing day, digital health tools are becoming an increasingly integral part of our daily routine, and are expected to play an even bigger role in the future. These tools are transforming how individuals maintain their well-being by providing convenient and accessible ways to track and manage their health. Mobile health apps are one such tool, which

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make it easy to monitor fitness progress, set and achieve goals, and track calorie intake. Additionally, wearable devices such as smartwatches and fitness trackers constantly monitor heart rate and sleep patterns, providing users with valuable insights into their overall health. Another important tool in the digital health toolkit is the use of smart home devices. These devices can remind individuals to take their medication on time, helping them to stay on top of their health and avoid missing important doses. A key driving force behind the evolution of digital healthcare is the widespread adoption of Internet of Medical Things (IoMT) devices [1]. These devices are equipped with sensors that can collect and transmit vital health data, allowing healthcare professionals to monitor patients remotely and provide more personalized care. As more and more people embrace these technologies, the potential for improved health outcomes and better quality of life becomes increasingly possible. These interconnected devices, from wearable sensors to remote monitoring systems, have revolutionized the healthcare landscape by generating unprecedented data. The continuous streams of real-time data collected from IoMT devices offer valuable insights into individuals' health statuses, enabling personalized and proactive care. The rise of IoMT devices has significantly contributed to the performance of artificial intelligence (AI) in healthcare applications. Fueled by the vast amount of healthcare data, AI-powered models show remarkable promise in disease diagnosis, treatment recommendations, and drug discovery [2]. These models analyze extensive healthcare data, uncover hidden patterns, and generate accurate predictions, ultimately improving patient outcomes and enhancing healthcare delivery.

However, the traditional method of training AI models in healthcare involves consolidating the data onto a single server, which brings various challenges. Healthcare data contains highly sensitive and confidential information, requiring strict privacy regulations to safeguard patients' rights. Centralizing the data increases the likelihood of unauthorized access, breaches, and potential violations of privacy regulations like HIPPA [3]. Moreover, the centralized approach introduces issues related to latency. When large amounts of data from IoMT devices are transmitted to a central server, it can lead to network congestion, delays, and higher communication costs. In healthcare, where making timely decisions is crucial, latency can significantly impact the effectiveness and efficiency of AI-powered applications. Furthermore, as the volume of healthcare data grows exponentially, the centralized training approach becomes less scalable. Storing and processing all the data in a single server becomes impractical and resource-intensive, making implementing AI-powered medical applications widely difficult. This abundance of data is valuable for training machine learning models, helping us maintain good health and prevent diseases.

Federated learning (FL) [4] is an approach introduced by Google to tackle privacy concerns and other challenges present in traditional centralized learning methods. Unlike centralized approaches, FL distributes the training tasks across multiple devices, ensuring private data remains on those devices. The FL process requires multiple rounds of communication until the model reaches the desired level of accuracy. In the context of healthcare applications, FL offers a solution to safeguard sensitive patient data. By keeping the data on local devices, FL minimizes the risks associated with centralizing and transmitting sensitive information. This decentralized approach ensures that personal health data remains secure and private. FL in healthcare faces challenges in privacy, ethical, and legal considerations. Privacy and security are the primary concerns to prevent malicious nodes from attacking the model. Healthcare providers who cooperate understand patient care extends beyond themselves. Lucchi et al. [5] is an advanced model for cooperation. Additionally, FL addresses the latency issues often encountered with centralized training [6]. By performing computations locally on edge devices, FL reduces the need for constant communication with a central server.

This reduces communication overhead and enables real-time decision-making, crucial in healthcare settings where timely actions can significantly impact patient outcomes.

In healthcare applications that utilize Federated Learning (FL) with numerous IoMT client devices, it becomes impractical to involve all clients in every training and model aggregation process due to limitations in communication and computing resources. During each round of FL, as illustrated in Fig. 1, a subset of clients is selected randomly [4] or with some level of randomness [7]. However, this selection approach raises concerns when certain clients, particularly those with limited resources like IoMT devices, experience longer processing times and potential failures in completing the training tasks. However, in the IoMT environment, where data can be very different from one device to another, this method cannot accurately select the devices that would be most helpful in speeding up the training process. These challenges significantly impact the accuracy of the trained models, ultimately affecting their effectiveness in healthcare settings [6]. Their location and availability influence the participation of clients in FL. Studies have revealed that the number of clients engaging in FL tends to vary depending on the time of day, with higher participation observed during nighttime compared to the daytime [8]. Consequently, relying solely on random client selection can result in fewer updates sent by clients, leading to the discarding of some FL rounds. This poses a challenge in effectively utilizing client data for model training in healthcare applications. It is worth mentioning that the current methods used in FL for healthcare do not fully tackle the issue of selecting suitable clients, considering factors such as their location, available resources, and device compatibility during the training phase [9]. Addressing these considerations is crucial to ensure that FL models in healthcare are trained efficiently, leveraging diverse client data while accommodating the limitations and characteristics of the participating devices.

To address the islanding data problem in digital healthcare applications, FL can be used by selecting appropriate clients. Deep learning (DL) algorithms are applicable in the context of FL for healthcare settings.

This paper proposes a novel client selection strategy for FL in healthcare. Instead of involving all devices in every training round, it sees each round as an opportunity to select

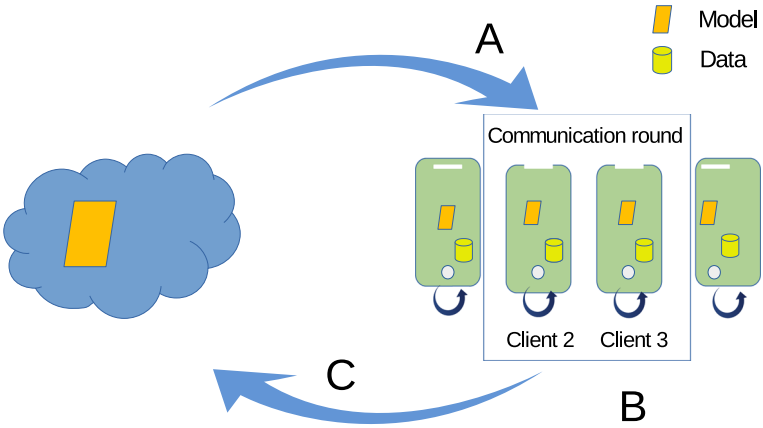


Fig. 1 Random client selection in FL process

the most impactful clients for efficient and effective model training. This strategy is motivated by Active learning [10] and the POWER-OF-CHOICE [11] strategy. In this new approach, clients are not chosen randomly for each round. Instead, they are chosen based on a probability that considers the quality of their current model and the amount of loss they have experienced. This means that clients with higher loss values are more likely to be chosen, as they will likely have more informative data. By utilizing active learning and prioritizing uncertainty estimation and client ranking, this approach can improve the performance of FL in domains such as healthcare. The system can prioritize clients who provide challenging or informative data, ultimately leading to better results.

In summary, this paper makes the following contributions:

- **Proposed Smart Client Selection Strategy**: In this paper, a novel method for FL is suggested, which views every training round as a process of selecting clients. The technique initially calculates an uncertainty score for each client, indicating the significance of their data in the overall model. Subsequently, the clients are ranked according to their respective uncertainty scores.
- **Enhanced FL Client Selection**: The proposed strategy for client selection involves identifying clients from a ranked list based on their uncertainty values. Higher loss value clients are more likely to take part in the training. Additionally, this approach guarantees the involvement of additional clients who might not be chosen in each round but can still contribute to the model. This ultimately leads to a better training effect on the final model.
- **Performance and Evaluation**: Through experiments, it has been shown that the suggested approaches have successfully attained the desired level of accuracy in the multi-label human activity classification task while using 30% fewer communication rounds and speeding up 1.3x the process.

This paper's sections are organized as follows: Section 2 summarizes related work, presenting an overview of the existing research and contributions in the field. Section 3 presents a detailed description of the Smart Client Selection strategies for the FL framework. This section comprehensively explains these strategies, highlighting their key features and advantages. Moving forward, Section 4 presents and analyzes the experimental results obtained from the study. The results are carefully examined and interpreted, shedding light on the performance and effectiveness of the proposed strategies. Finally, in Section 5, the paper concludes by summarizing the main findings and implications of the work.

2 Related works

This section reviews the literature related to this proposed work in two broad categories: FL in healthcare and its challenges and client selection strategies in FL.

2.1 Federated learning in healthcare and its challenges

Federated Learning (FL) is a machine learning technique that allows the development of AI models using data from various sources while ensuring the privacy of data. This involves training the model on decentralized edge devices that store local data samples privately. The healthcare industry is increasingly interested in this approach as it enables the creation of AI models using sensitive health data without violating patient privacy. McMahan et al [4]. presented a practical FL system that enables model training using sensitive data on edge

devices while preserving user privacy. This method inspired many subsequent works on privacy-preserving federated learning for healthcare. Recent studies have focused on applying FL to train models on sensitive health data. For example, Jie Xu et al [12]. proposed an FL framework to develop a pneumonia diagnosis model from chest X-ray images across multiple hospitals. Their approach showed comparable performance to centralized learning while keeping the data decentralized. Similarly, Mohan et al [13]. presented an FL architecture to detect diabetic retinopathy from retinal fundus images. Their method achieved high accuracy while protecting patient privacy. FedHealth [14] and FedHome [15] are two proposed frameworks for FL in healthcare and smart homes, respectively. They aim to enable privacy-preserving analytics from distributed data sources like wearable, mobile apps, and smart home sensors. FedCure [16] and FedRH [17] introduced personalization in FL for healthcare applications to address model heterogeneity but required an efficient client selection strategy to address issues with system heterogeneity. The model in this study [18] was trained on data from multiple hospitals, which allowed for a more comprehensive understanding of the disease. This is because the data from different hospitals can be different, and by training on data from multiple hospitals, the model can learn to account for these differences. The study results showed that the model could accurately predict clinical outcomes in COVID-19 patients. This indicates that FL is a potentially effective method for developing machine learning models for healthcare applications that can aid in preserving patient privacy. A major challenge in FL for healthcare is dealing with data heterogeneity across sources [19]. Hospitals may have different population characteristics, imaging equipment, and annotation standards. This can reduce model accuracy if not addressed properly. Proposed solutions include: using robust loss functions, data augmentation, and federated transfer learning [20]. The selection of data sources (clients) to take part in each FL round is a crucial factor to consider in these systems. Random client selection is a simple approach but may impact model accuracy in some cases: data heterogeneity, limited communication, and high client dropout [9]. To summarize, FL is a promising methodology for distributed healthcare ML. However, it still faces significant challenges such as data heterogeneity, client selection strategies, and privacy concerns. These challenges require further research. FL has the potential to enable large-scale healthcare analytics while safeguarding sensitive data with appropriate solutions.

2.2 Client selection strategies for federated learning

The diverse communication environments and varying computation resources among clients can challenge the overall training speed in FL for healthcare applications. Researchers have explored methods to make intelligent decisions regarding client selection to overcome this client heterogeneity and expedite the convergence speed [9]. The aim is to reduce the burden of communication and enhance the effectiveness of FL in healthcare settings. This can be achieved by selecting clients based on their abilities and resources, optimizing the training process, and achieving faster convergence. As a result, the performance of FL models in healthcare applications can be improved significantly. Wang *et al.* [21] introduced an interesting approach involving the Federated Deletion and Shap methods. These methods were designed to measure the contributions made by multiple parties in both horizontal and vertical FL models. Even though this method requires a lot of computing power, it ensures a just assessment of every client's input. Nishio *et al.* [7] involved selecting as many clients as possible who possess abundant data, even in complex communication environments. These approaches focus on enhancing the client selection process and boosting the performance

of FL models in diverse FL scenarios. Luping *et al.* [22] used knowledge of the overall model to determine if their modifications were in line with the cooperative aspect of the training procedure. Clients can actively participate in global aggregation by picking the most advantageous aspects of their updates, lessening communication burden and hastening convergence. Nagalapatti *et al.* [23] examined client variations using game theory concepts. They identified the Shapley Value for each client through their analysis, and they then used it to establish how important each client was to the server's learning objectives. These innovative approaches contribute to optimizing client selection in Federated Learning, enhancing the overall efficiency and effectiveness of the learning process. Cho *et al.* [11] discovered that choosing clients with greater local loss can accelerate the convergence process, indicating a higher potential for model improvement. Their proposed POWER-OF-CHOICE algorithm can yield up to 3X faster convergence and 10% higher test accuracy than conventional federated learning with random client selection. Most client selection methods discussed earlier have a common drawback: they require training all clients upfront, leading to a considerable waste of computing resources.

In summary, while random client selection is simple, more intelligent techniques can help improve the performance of FL in healthcare applications. A balanced approach incorporating randomness and data-driven client preferences may achieve the best results. However, this paper introduces a novel client selection approach that overcomes this limitation. By implementing this strategy, the issue of inefficient resource utilization during client training can be effectively addressed. The proposed approach offers a promising solution to optimize the client selection process in Federated Learning, enhancing the system's overall efficiency.

3 Methodology

This paper considered a horizontal FL setting [4]. Every client possesses identical features while having distinct samples of data. In the healthcare context, this paper explores the use of the FL framework. This framework involves a central server, referred to as C_s. Improving the performance of a model is crucial and requires playing a significant role. The server achieves this by involving various entities represented by the set $C_s = \{1, 2, 3, ..., n\}$ in the training process. These entities could be hospitals, clinics, or other relevant healthcare stakeholders. By encouraging their participation, the server aims to enhance the overall quality and effectiveness of the model with smart client selection strategies. Each client in the system possesses its local dataset, which serves as a source of training data for the model. The use of these devices is hindered by challenges related to the time and energy required for training. To predict y_i based on x_i. This study will assume that each healthcare edge device, designated by i, has labeled data (x_i, y_i) and a model with parameters $w_i \in \mathbb{R}^d$. $\hat{y}_i = f(x_i; w_i)$ stands for the expected value. The model w_i is trained by minimizing a loss function $l(x_i, y_i; w_i)$, where $w_i(t)$ reflects the model parameters' values at training iteration t.

3.1 Federated learning

The concept of FL enables healthcare providers to train a AI model without violating the privacy of patient data. This process involves a cloud FL server and healthcare edge devices, as illustrated in Fig. 2. All participating clients (edge devices) receive the global model from the FL servers. Each edge device trains its model using local data, such as medical records of individual patients. This process continues till the model converges or obtains the desired

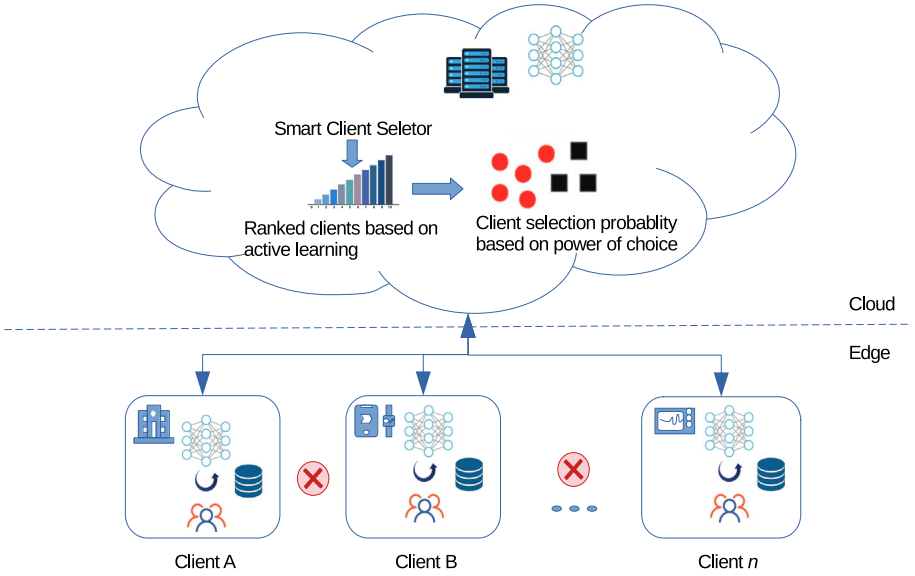


Fig. 2 The proposed Smart Client Selection Strategy for FL in healthcare involves the selection of devices (clients) that participate in the FL process using Active Learning with Power of Choice methods

accuracy with t no of communication rounds. This paper takes multi-label classification tasks as a problem and adopts deep neural networks (DNN) [24] in both Cloud and edge models for learning. Deep networks train classifiers and effectively learn complex features from raw user data, carrying out more complex computing tasks. The cloud model is denoted as f_C, and the learning objective function becomes:

$$\arg\min_{\Theta_C} \mathcal{L} = \sum_{i=1}^{n} \ell\left(y_i, f_C\left(\mathbf{x}_i\right)\right) \tag{1}$$

The metric used to evaluate a deep learning model's performance during the training process is known as the network loss, denoted as $\ell\left(y_i, f_C\left(\mathbf{x}_i\right)\right)$. It measures how well the model's predictions differ from the actual outcome it's trying to learn. To understand this better, let's consider the example of teaching someone to distinguish between different types of birds. If show them pictures of various birds and ask them to identify them, their guess may be different from the correct answer. The difference between their guess and the correct answer is similar to the network loss, which represents the deviation of the model's predictions from the actual outcome. For illustration, the cross-entropy loss might be the purpose of classification tasks. $\{x_i, y_i\}_{i=1}^{n}$ are samples available from datasets, the parameters to be learned, or the weights and bias in DNN, are indicated by the symbol Θ. The model will be transmitted to all edge devices by the cloud server. Participated clients trained local model f_E using local data. $\left\{x_i^E, y_i^E\right\}_{i=1}^{n^E}$ denotes samples from local datasets E, and the learning objective function for the client becomes:

$$\arg\min_{\Theta_E} \mathcal{L}_1 = \sum_{i=1}^{n^E} \ell\left(y_i^E, f_E\left(x_i^E\right)\right) \tag{2}$$

After training the local edge model with their local data, client devices transmit model weights back to the FL cloud server. The server aggregates the model based on FederatedAverage (FedAvg) algorithm [4], and the averages are calculated using the following:

$$f'_C(w) = \frac{1}{K} \sum_{k=1}^{n} f_{E_k}(w) \tag{3}$$

Here, $f'_C(w)$ denotes the updated cloud model, and $f_{E_k}(w)$ represents the model of the k-th edge device. For Example, A FL scenario where five edge devices (denoted as E_1, E_2, E_3, E_4, and E_5) train a global cloud model for a medical diagnosis task. Each edge device trains its local model using its own local dataset (which might vary due to patient demographics, diseases prevalent, etc.). After training their local models, each edge device sends its model updates (i.e., model parameters) back to the central server for aggregation. The server then computes the updated global model using the FedAvg algorithm, as described in the (3).

The conventional Federated Learning (FL) method randomly selects a subset of devices for every training iteration. However, this approach can be inefficient in certain cases. For example, choosing devices that have low-quality data or are located in remote areas with poor internet connectivity can hinder the training process and lead to a suboptimal model. Additionally, there may be high transmission costs between the cloud server and the devices [11]. To overcome these issues, a new smart client selection stratergy has been proposed in this paper that employs active learning principles to strategically select the devices. This approach minimizes the number of training sessions required to achieve the desired accuracy level. The proposed method has the potential to enhance the efficiency and effectiveness of the FL process by selecting the most suitable devices for collaboration while also minimizing transmission costs and training time.

The overall learning process of the proposed framework is as follows.

1. The cloud FL server C_s initially sends the global model to all participating clients n.
2. Each client uses local data to train the global model and a validation dataset or a representative subset of client data to assess the global model.
3. Each client calculates the uncertainty estimation using BALD [25], which involves two steps: Obtain predictions from the global model for multiple samples with dropout enabled, and calculate the average entropy of predictions across the dropout samples.
4. All participated clients return to the updated model and uncertainty values which measures how much the client's data contributes to the overall model.
5. FL server aggregates the client's weighs using FedAvg [4], Smart Client Selector sort the clients based on their uncertainty values in descending order and selects the clients for the next communication round based on Power of Choice client selection [11].
6. A smart client selection approach is used by FL server to send the updated global model to a selected chosen clients.

From steps 2 to 6, continue until convergence or a certain number of communication rounds.

3.2 Bayesian information theoretic active learning with POWER of CHOICE

Active Federated Learning (AFL) is a novel method that focuses on training the model on devices that can provide more valuable information during a specific training cycle. This approach is driven by active learning, a machine-learning method that identifies the most useful data points to label. By using the POWER of CHOICE client selection strategy, the model's efficiency can be improved, and convergence time can be reduced [11]. In AFL,

each participating device calculates a value function $V : X \times Y \times \mathbb{R}^d \rightarrow \mathbb{R}$ to evaluate the potential benefit of training on that particular device. Here, X represents the input space consisting of the data that the model processes, while Y denotes the output space containing the corresponding labels or predictions associated with the input data. Then, they send their value functions score $v_i \in \mathbb{R}$ with training losses to the cloud server, which collects the scores and ranks the clients based on that. Although the score depends on the model parameters, propagating the model is expensive; thus, the user only receives a fresh score during the initial training of the client device. Smart client selector will generate the client probability from ranked clients based on their training losses. The cloud server then uses these probabilities to select the next cohort of devices to train on.

This paper used Bayesian Active Learning by Disagreement (BALD) [25] to conduct active federated learning. With a Bayesian method, BALD models the correlation between inputs x and outputs y using latent factors z. As measured by Shannon's entropy, reducing the number of potential hypotheses and minimizing parameter uncertainty are the main objectives of information-theoretic active learning.

$$D_i^* = \arg\min_{D_i} H\left[z \mid D_i\right] = -\int p\left(z \mid D_i\right) \log p\left(z \mid D_i\right) \mathrm{d}z, \tag{4}$$

Here, D_i^* represents the optimal selection of training data D_i for edge device i that minimizes the conditional entropy $H\left[z \mid D_i\right]$. This implies finding the subset of training data that reduces uncertainty about the latent factors z the most. Therefore, the equation searches for a data point (D_i^*) that leads to the largest decrease in the average uncertainty about the latent factors (z) when added to the existing data. In other words, we aim to minimize uncertainty about z by selecting the most informative data points. However, the equation itself does not represent the result of the optimization. It defines the function that needs to be optimized to find the optimal data selection D_i^*. For example, a box full of different colored balls, and each ball represents a data point. The latent factors could be the underlying properties, like size or material, that determine the ball's color. BALD helps you pick the "most informative" balls that reveal the most about these hidden properties, minimizing the overall uncertainty about how they influence the colors.

The given problem involves finding the data point, denoted as x_i, that can maximize the expected reduction in posterior entropy. This task is challenging as it deals with a posterior distribution, represented by $p\left(z \mid D_i\right)$, derived from a training set called D_i. Despite the problem's NP-hard nature, a myopic policy can still provide a near-optimal solution.

$$\arg\max_{x_i} H\left[z \mid D_i\right] - E_{y_i \sim p\left(y_i \mid x_i, D_i\right)} H\left[z \mid y_i, x_i, D_i\right]. \tag{5}$$

The objective can be restated as calculating the entropy in the y_i space.

$$\arg\max_{x_i} H\left[y_i \mid x_i, D_i\right] - E_{z \sim p\left(z \mid D_i\right)} H\left[y_i \mid x_i, z\right], \tag{6}$$

This paper used the uncertainty measure called BALD to guide our decision-making. By calculating the mutual information between predictions and the posterior distribution of the model, the BALD technique calculates the amount of information received about the model parameters. It helps us identify data points with the most disagreement among the model parameters, indicating higher uncertainty. The following equation can summarize the calculation formula:

$$V_i\left(y_i, z \mid x_i, D_i\right) = H\left[y_i \mid x_i, D_i\right] - E_{p\left(z \mid D_i\right.}\left[H\left[y_i \mid x_i, z\right]\right]. \tag{7}$$

When implementing dropout, it can be viewed as a way to approximate variational Bayesian methods [26]. These techniques estimate intractable integrals that arise in Bayesian inference, mainly in complex statistical models. For each sample, this strategy needs to produce numerous forecasts with dropouts. To compute, the average of the entropies of the individual forecasts is subtracted from the entropy of the average of these predictions. A higher value means that the sample offers more informative insights into the current model, leading to improved performance. The Smart client selector in the cloud server will rank clients based on this value.

This paper adopts the POWER of CHOICE [11] client selection strategy π_{loss}. This strategy selects clients with higher loss function values for training from ranked clients by smart client selector to accelerate convergence speed and incorporate valuable insights from those clients. The π_{loss} strategy incorporates the importance weight of client v_k and a selection probability function. This strategy quantifies the potential value of client k in accelerating the model's convergence. For instance, in a classification task using the cross-entropy function, the importance weight of client k is defined as follows:

$$v_k^{(t+1)} = \begin{cases} \ell_k^{(t)}(w) = \sum_{i=1}^{c} q(y=i)\mathbb{E}_{x|y=i}\left[\log f_i(x,w)\right], & \text{if } k \in U^t \\ v_k^{(t)}, & \text{else,} \end{cases} \tag{8}$$

Where k represents the client and U^t represents the set of selected clients in round t.

Due to communication and hardware constraints, it is difficult for the server to obtain loss values from all clients in FL. Therefore, for clients that did not participate in the current round of training, their client value for the next round remains the same as the previous round, that is, $v_k^{(t+1)} = v_k^t$. Although this approach may introduce outdated information, it has a negligible impact on the selection performance. This is because the loss values of clients generally decrease as the number of training rounds increases. When a newly selected client exhibits a lower loss value, its selection probability is reduced. This creates greater chances for other clients not chosen in the current round to be selected in the next round. Because of this, the global model may gain from a greater variety of information and instruct from a wider group of clients. As shown in Fig. 2 smart client selector generated the ranked clients based on the uncertainty value and used loss values to convert the probability of client selection, FL cloud server selected the sub-set of clients based on this new client selection strategy in each communication rounds that will help to improve the global model performance, reduce the communication and computation overhead.

4 Experiments

This section comprehensively tests the proposed smart client selection strategies for human activity recognition (HAR) classification tasks.

4.1 Dataset

For experiments, utilized the publicly available MHEALTH dataset [27] for human activity recognition. This dataset consists of recordings from 10 volunteers engaging in 12 different physical activities while wearing sensors on their right wrist, chest, and left ankle. The sensors capture data from accelerometers, gyroscopes, and magnetic sensors, providing information about body movements and vital signs. Additionally, the chest sensor records 2-lead ECG

readings, which can be useful for cardiac monitoring and analyzing the impact of the activity on the ECG. The dataset includes 23 sensor recordings, combining acceleration, turn rate, magnetic field orientations, and ECG data. It comprises 12 unbalanced classes and a total of 12,15,746 samples.

4.2 Implementation details

In this study, the PFL platform[1] was utilized as a baseline federated learning (FL) framework to implement the proposed smart client selection strategies. The implementation was done using the latest version of Python, and the dataset was divided into different clients using the same platform. The proposed smart client selection strategy was evaluated in the context of the human activity recognition (HAR) classification task within the FL setting using the MHEALTH dataset. The dataset was distributed among ten clients in three distinct scenarios: IID balanced, IID unbalanced, and label skew-based pathological Non-IID. Each experiment was carried out separately, allowing for a comprehensive evaluation. Moreover, additional experiments were performed to assess the performance in scenarios with system heterogeneity, including dropout clients, slow learners, and slow senders. The proposed CNN+LSTM architecture was adopted, comprising two 1D convolution blocks with 32 and 64 filters, respectively. These blocks employed three kernels to convolve the input data. Batch normalization layers were incorporated in both blocks to normalize and reconstruct each batch of training data, thereby enhancing training speed and accuracy. An activation function called Rectified Linear Unit (ReLU) was used to calculate feature maps. These feature maps were subsequently downsampled using a max-pooling layer. The output obtained from the convolutional layers was then inputted into a single LSTM block comprising 64 memory cells, with the hyperbolic tangent (tanh) activation function applied. A dense layer with ReLU activation function was used, followed by another dense layer with softmax activation function serving as the classification layer. For the experiments, the local epoch was set to E = 5, indicating the number of passes through the local dataset during training. The batch size, denoted as B, was set to 10, representing the number of samples processed together during each iteration. To facilitate the training process, a learning rate of 0.05 was chosen.

4.3 Evaluation metrics and baseline

In the conducted experiment, a conventional FL setup was employed, which represents a decentralized learning approach that facilitates the training of DL models on decentralized data sources without data transfer to a central location. The selection of clients was conducted randomly to participate in the FL process, and each client trained the machine learning model using its own local data. To evaluate the performance of the algorithms, the primary focus was on a metric called convergence speed, which measures the number of rounds required for each method to achieve the target accuracy. This metric reflects the speed of convergence of the machine learning model to a good solution. The main aim of the experiment was to establish a baseline for the performance of conventional FL setups with random client selection, which can be utilized to compare with the proposed smart client selection strategy.

[1] https://github.com/TsingZ0/PFL-Non-IID

4.4 Results and discussion

In this study, we conducted experiments using a standard Federated Learning (FL) setup, where clients were randomly selected to participate. The primary metric used to evaluate the performance of different strategies was the convergence speed, which was measured by the number of rounds required for each method to achieve the target accuracy. This allowed us to assess and compare the efficiency of the strategies in reaching the desired performance level. In our study, we examined the efficiency of proposed smart client selection strategies using two distinct scenarios and 10 FL clients. We used a dataset consisting of recordings from 10 volunteers who performed 12 different physical activities. This approach allowed us to distribute individual volunteer data to each client, making the client datasets more diverse and representative in our FL experiments. The first scenario involved balanced and unbalanced data distribution, with ten clients and a desired target accuracy of 96% for balanced data and 95% for unbalanced data. In each round, at least two clients were required to participate in the FL process. For the IID-balanced scenario, random client selection took 100 communication rounds to achieve the target accuracy, while the proposed approach achieved it within 43 communication rounds. This was done in half the time taken by the random selection approach, with an overall F1 score of 96%. In the IID-Unbalanced scenario, each client had a different amount of data compared to the balanced scenario, where each client contained an equal number of samples. The target classification accuracy was 95%. To achieve this, the random client selection method took 100 communication rounds, while the proposed method reached the target within 54 communication rounds. For the non-IID scenario, the target accuracy was 75%. The random strategy took 100 communication rounds to achieve this, while the proposed approach achieved it within 62 communication rounds.

The results of IID-balanced and unbalanced distribution are shown in Figs. 3 and 4, respectively. When compared to the outcomes of a random client selection, the proposed techniques were more accurate and required fewer communication rounds. In an unbalanced scenario some clients have more data so in random clients selection initial few rounds randomly selected clients with fewer data, so the performance fluctuates in a few rounds, but the proposed strategy clearly selected clients with more data so performance improved compared to random selection. It also shows that it takes fewer communication rounds in both scenarios and Improves the model performance.

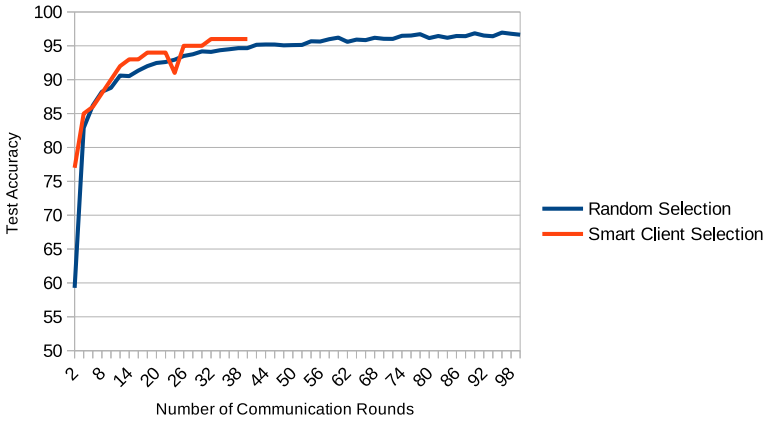


Fig. 3 Test Accuracy vs. Communication Rounds on IID Balanced Distribution

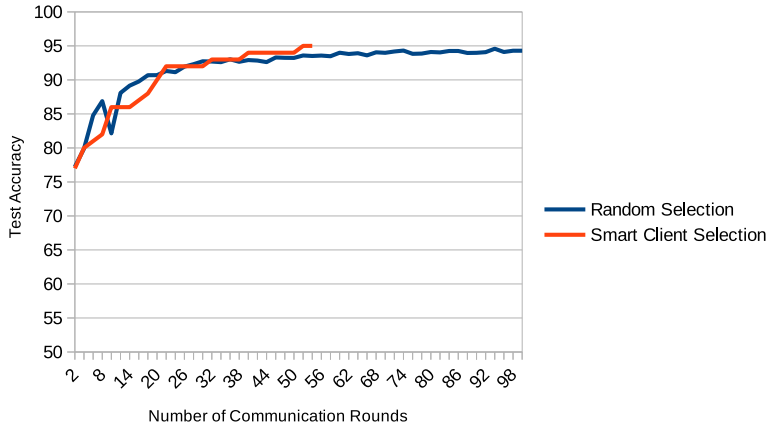


Fig. 4 Test Accuracy vs. Communication Rounds on IID Unbalanced Distribution

The Second scenario tested on label skewed pathological Non-IID distribution to show a data heterogeneity environment. In this experiment, each client contains the data with different labels. For example, a few clients contain only two labels, and others contain more. Additionally, practical FL scenarios like client dropout, slow learners, and slow senders if the server selects the clients for FL process but some issues like network and power they unable to join and perform on time. The 10% selected clients will randomly drop at each training round, and 10% of clients are slow learners and senders. Figure 5 illustrates the performance in a heterogeneous environment. This particular scenario requires more communication rounds to achieve the target accuracy compared to the cases of IID-balanced and unbalanced data distributions. During the initial stage of this scenario, improving the model performance takes more rounds due to the heterogeneity of the data and challenges related to client speed and data drift. However, even in this challenging setting, the proposed strategy outperforms communication rounds and model performance. The results in Table 1 demonstrate the proposed strategy's effectiveness in reducing the training process and achieving improved model performance. The client selection mechanism is based on a probability that

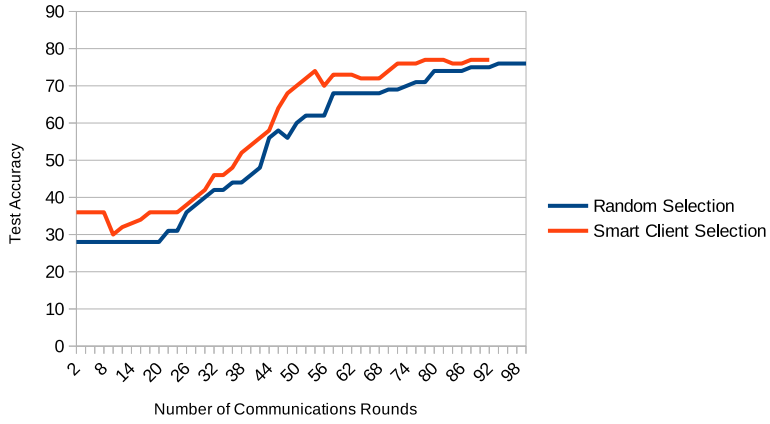


Fig. 5 Test Accuracy vs. Communication Rounds on pathological Non-IID distribution with Pathological FL scenario

Table 1 Statistics of communication rounds and time (seconds) taken for FL training

Environment	Strategy	Target Communication Rounds	Time Taken (Sec)	Speed up
IID-Balanced (Target Accuracy 96%)	Random	100	46.91	1.0x
	UCS-FL [28]	43	20.18	1.6x
	Proposed	43	21.68	1.6x
IID-Unbalanced (Target Accuracy 95%)	Random	100	48.32	1.0x
	UCS-FL [28]	72	32.98	1.3x
	Proposed	54	23.12	1.5x
Pathological Non-IID with Practical FL (Target Accuracy 75%)	Random	100	85.63	1.0x
	UCS-FL [28]	91	71.52	1.3x
	Proposed	68	61.21	1.4x

considers the quality of the client's current model and the magnitude of the loss they have experienced. Consequently, clients with higher loss values are more likely to be selected as they are expected to possess more informative data. This approach integrates active learning principles focusing on uncertainty estimation and client ranking. The system can prioritize clients that contribute challenging or informative data by doing so. The experiments show efficient client selection in FL significantly enhances the performance of healthcare applications in IoMT environments. In this example, take a look at how AI can be used in remote patient monitoring systems that are deployed in IMoT environments. In such systems, wearable devices are used to collect physiological data continuously from patients, including their heart rate, blood pressure, and activity levels. FL can be used to develop AI models that can analyze this data in real time. This analysis can help to detect health issues early and enable personalized healthcare interventions. The proposed Efficient client selection in FL allows for the inclusion of diverse data sources from IMoT devices while mitigating communication and computational constraints. For example, by prioritizing clients with high-quality data and reliable connectivity, the FL system can ensure timely updates to the AI model without compromising accuracy or efficiency.

Overall, efficient client selection in FL optimizes the utilization of data from IMoT devices, leading to improved performance and effectiveness of healthcare applications. By leveraging case studies like remote patient monitoring and disease prediction, This paper demonstrates the tangible benefits of FL in enhancing healthcare outcomes in IMoT environments.

5 Conclusion

This paper introduces a smart client selection strategy for healthcare applications in federated learning, combining active learning and power of choice. The strategy aims to enhance model performance while minimizing communication rounds. Participating clients receive

the global model, train it using local healthcare data, and calculate a value function indicating their training's potential benefits. These value functions and training losses are sent to the cloud FL server. A smart client selector collects and ranks the clients based on these scores, generating client probabilities accordingly. Experimental results demonstrate the strategy's effectiveness in accelerating training, reducing communication costs, saving computing resources, and improving model performance. Future research directions can focus on specialized uncertainty calculation methods for IoMT environments and mitigating the impact of loss-based client selection on model precision and accuracy in heterogeneous data settings. Developing adaptive and robust client selection mechanisms considering data diversity can enhance model performance further. Pilot studies or clinical trials in real-world healthcare settings could shed light on the proposed client selection techniques' practicality, effectiveness, and scalability.

Data Availability Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Conflicts of interest The authors declared that they have no conflicts of interest in this work.

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